

Unless specified otherwise, assume the working fluid is air and it can be treated as an ideal gas with constant specific heats $\gamma=1.4$. The specific gas constant for air is:

$$R = 287 \text{ J/(kg} \cdot \text{K)} \text{ in SI units or } R = 1716 \text{ ft} \cdot \text{lbf / (slug} \cdot \text{R)} \text{ in English units.}$$

In solving the problems below, unless otherwise specified, you may either use the equations derived in the text or the course notes or the tables in the text, or the online compressible calculator <https://devenport.aoe.vt.edu/aoe3114/calc.html> . Make it clear which you are using and, if you use the tables, use linear interpolation for values of the parameters that do not correspond to entries in the table.

1. Consider a nozzle with $A_e/A_{th} = 3$, reservoir pressure $p_{res} = 12 \text{ atm}$, and temperature $T_{res} = 500 \text{ K}$. Calculate the Mach number, pressure and temperature at the throat and the exit for the following cases:

- (a) The flow is supersonic at the exit (perfectly expanded flow).
- (b) The flow is subsonic for the entire nozzle except at the throat where it is sonic (choked subsonic flow $M_{throat} = 1$)
- (c) Calculate the required reservoir pressure for choked subsonic flow when the exit pressure is $p_{exit} = 0.468 \text{ atm}$.

2. A nozzle is designed to expand air from a chamber in which the pressure and temperature are 1000 KPa and 60°C respectively to a Mach number of 3.5. The throat area of the nozzle is 0.05m². Find

- a. The exit area of the nozzle
- b. The mass flow rate through the nozzle when operating at design conditions (supersonic exit)
- c. The back pressure for which it will be a normal shock wave on the exit plane of the nozzle
- d. The range of back pressures where there will be a normal shock wave in the nozzle
- e. The range of back pressures over which there will be oblique shocks at the exit (overexpanded nozzle)
- f. The range of pressures over which there will be expansion waves outside the nozzle (under-expanded nozzle)

3. Consider flow through a converging-diverging nozzle with an exit-to-throat area ratio of 1.3. The exit has a diameter of 0.75 m. The upstream value of the stagnation temperature is $T_t = 900 \text{ K}$. The upstream value of the stagnation pressure is P_t , which is varied for parts (a)-(d) below. The back pressure (atmospheric pressure) is $P_b = 1 \text{ atm}$. For each of the cases below, (a)-(d), classify each of the conditions as one of the following:

- (a) $P_t = 1.25 \text{ atm}$ (b), $P_t = 1.45 \text{ atm}$, (c) $P_t = 1.65 \text{ atm}$, (d) $P_t = 3.30 \text{ atm}$,
 - (i) Subsonic isentropic flow throughout with subsonic conditions at the throat
 - (ii) Subsonic isentropic flow throughout with sonic conditions at the throat
 - (iii) Shock inside the nozzle

- (iv) Shock at the nozzle exit
- (v) Overexpanded flow with shocks outside the nozzle
- (vi) Perfectly expanded supersonic flow
- (vii) Underexpanded flow with expansion waves outside the nozzle.

Also, determine or do the following for each case, (a)-(d):

- The exit Mach number, M_e .
- The mass flow rate through the nozzle, $\dot{m}$.
- The thrust generated by the nozzle, which is given by
- The nondimensional thrust coefficient, $C_T = \text{Thrust}/(P_t \cdot A_{th})$.
- Plot the location of each case, (a)-(d), on the Nozzle Operating Map, marking the locations as (a), (b), (c) or (d).

